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DS 120 Pop Quiz 3

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Instructions

- Unless otherwise instructed by a Teaching Fellow, the time limit for this quiz is **10 minutes**. Please manage your time accordingly.
- At the end of the quiz, please **submit** your completed paper to your Teaching Fellow.
- Only tools required for writing your answers are allowed on the desk. All other items must be put away.

Section 1 - Multiple Choice

Select only one option

Question 1. Let

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}.$$

Which of the following statements is true?

- (A) A is symmetric and B is symmetric.
- (B) A is symmetric and B is skew-symmetric.
- (C) A is skew-symmetric and B is symmetric.
- (D) Both A and B are skew-symmetric.

Question 2. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

The rank of A is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Section 2

Find the line of best fit of the form

$$y = mx + b$$

for the following data points: $(-1, 2), (0, 0), (1, 4), (3, 5)$. Use the least squares method to determine the values of m and b .

$$m = 2, b = 0$$